Minime CI Workflow Architecture (Vertical Pipeline)

Unified GitHub Actions Build System — .github/workflows/build.yml **[Shared]**
Legend: **[Shared]** = Dual-distro / shared across targets | **[Specific]** = Single-distro, board, or libc specific

1. Trigger / Entrypoint

- **Workflow File:** .github/workflows/build.yml **[Shared]**
- **Triggers:** Push to main, Pull Request, or workflow_dispatch

↓ *(Triggers parallel Tier 1 jobs)*

2. Parallel Tier 1 Builds (Bootloader & UIs)

2A. Bootloader Build

build-bootloader (AMD64)
- **Workflow Job:** build-bootloader **[Shared]**
- **Runner:** ubuntu-latest
- **Container:** minime-glibc:latest **[Shared]**
- **Script:** scripts/build-bootloader.sh **[Shared]**
- **Cache Key:** bootloader-* (hash of uboot/**)
- **Configs & Patches:**
  - minime/uboot/config/uboot.config **[Shared]**
  - minime/uboot/config/ddr3.defconfig **[Specific]** (H700)
  - minime/uboot/out/rk3566/rkbin/ **[Specific]** (RK3566)
- **GH Artifact:** bootloader-out

2B. UI Build (musl)

build-ui (musl) (ARM64)
- **Workflow Job:** build-ui (musl) **[Shared]**
- **Runner:** ubuntu-24.04-arm
- **Container:** minime-musl:latest **[Shared]**
- **Script:** scripts/build-ui.sh minui|allium musl **[Shared]**
- **Cache Key:** ui-musl-* (hash of ui/**)
- **Submodules:**
  - minime/ui/minui/ **[Shared]** (dual musl/glibc)
  - minime/ui/allium/ **[Shared]** (dual musl/glibc)
- **GH Artifact:** ui-musl
  - minui-musl-aarch64.tar.xz
  - allium-musl-aarch64.tar.xz

2C. UI Build (glibc)

build-ui (glibc) (AMD64)
- **Workflow Job:** build-ui (glibc) **[Shared]**
- **Runner:** ubuntu-latest
- **Container:** minime-glibc:latest **[Shared]**
- **Script:** scripts/build-ui.sh minui|allium glibc **[Shared]**
- **Cache Key:** ui-glibc-* (hash of ui/**)
- **Submodules:**
  - minime/ui/minui/ **[Shared]** (dual musl/glibc)
  - minime/ui/allium/ **[Shared]** (dual musl/glibc)
- **GH Artifact:** ui-glibc
  - minui-glibc-aarch64.tar.xz
  - allium-glibc-aarch64.tar.xz

↓ *(Uploads artifacts to GH & Passes to Tier 2)*

3. Artifact Transfer & Caching Layer

- **Actions Used:** actions/download-artifact@v4 **[Shared]**, actions/cache/restore@v4 **[Shared]**
- **Artifacts Downloaded:** bootloader-out + matching UI artifact (ui-musl for Alpine, ui-glibc for Buildroot)
- **Tier 1 Output Caches:** Bootloader Cache (bootloader-*), UI Cache (ui-musl-*, ui-glibc-*)
- **Tier 2 OS Caches:** CCache (~/alpine-ccache, ~/.buildroot-ccache) + DL Cache (~/alpine-dl, ~/.buildroot-dl) **[Specific]**

↓ *(Fans out to 6 parallel matrix jobs)*

4. Target Component Compilation (build.sh)

4A. Alpine Component Builder (musl)

Alpine (musl) Component Builder (3 Jobs)
- **Command:** make -C minime/targets/alpine components BOARD=<board> **[Specific]**
- **Builder Script:** minime/targets/alpine/scripts/build.sh **[Specific]**
- **Inputs:** minime/boards/<board>/ **[Specific]**, minime/boards/common/ **[Shared]**
- **Produced Component Artifacts:**
  - system.erofs (Immutable EROFS rootfs with Mesa/Panfrost)
  - Image (Linux kernel binary)
  - initramfs & *.dtb (Device Tree Blobs)

4B. Buildroot Component Builder (glibc)

Buildroot (glibc) Component Builder (3 Jobs)
- **Command:** make -C minime/targets/buildroot components BOARD=<board> **[Specific]**
- **Builder Script:** minime/targets/buildroot/scripts/build.sh **[Specific]**
- **Inputs:** minime/boards/<board>/ **[Specific]**, minime/boards/common/ **[Shared]**
- **Produced Component Artifacts:**
  - system.erofs (Immutable EROFS rootfs with glibc & Mali drivers)
  - Image (Linux kernel binary)
  - initramfs & *.dtb (Device Tree Blobs)

↓ *(Passes components (system.erofs, Image, DTBs) to Packaging)*

5. Image Assembly & Packaging (mkimage.sh / mkupdate.sh)

5A. Alpine Image Assembly (musl)

Alpine Image & Update Packager (musl)
- **Commands:**
  - make image BOARD=<board> UI=minui|allium **[Specific]**
  - make update BOARD=<board> UI=minui|allium **[Specific]**
- **Packaging Scripts:**
  - minime/build/genassets.sh **[Shared]**
  - minime/build/mkimage.sh **[Shared]**
  - minime/build/mkupdate.sh **[Shared]**
- **Produced Release Archives:**
  - *.img.xz (Raw SD card flashable image)
  - *.tar.xz (OTA update package)

5B. Buildroot Image Assembly (glibc)

Buildroot Image & Update Packager (glibc)
- **Commands:**
  - make image BOARD=<board> UI=minui|allium **[Specific]**
  - make update BOARD=<board> UI=minui|allium **[Specific]**
- **Packaging Scripts:**
  - minime/build/genassets.sh **[Shared]**
  - minime/build/mkimage.sh **[Shared]**
  - minime/build/mkupdate.sh **[Shared]**
- **Produced Release Archives:**
  - *.img.xz (Raw SD card flashable image)
  - *.tar.xz (OTA update package)

↓ *(Uploads output archives to GitHub Releases)*

6. Release & Delivery

- **Workflow Step:** Upload to Testing Release in .github/workflows/build.yml **[Shared]**
- **Tool / CLI:** gh release upload testing /*.img.xz /*.tar.xz --clobber **[Shared]**
- **Destination:** GitHub Releases — **testing** tag